

Markscheme

Mathematics: analysis and approaches
Practice question bank

124. (a)

A bag contains 12 balls: 5 red, 4 blue and 3 green. A selection of 5 balls is made.

Find the number of ways of selecting 5 balls such that at least 3 are red.

exactly 3 red: $\binom{5}{3} \binom{7}{2} = 10 \times 21 = 210$ *(M1)* A1

exactly 4 red: $\binom{5}{4} \binom{7}{1} = 5 \times 7 = 35$ A1

exactly 5 red: $\binom{5}{5} \binom{7}{0} = 1$ A1

sum: $210 + 35 + 1 = 246$ A1

[6 marks]